

Windows Event Logs, Analysis, Registry & File System Forensics

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When to perform: During/after host analysis for deeper reconstruction, timelines, and persistence details. **Why:** These artifacts survive reboots and provide reliable evidence of authentication, execution, file activity, and tampering.

Major Artifacts & What They Reveal

Artifact	Reveals	Common Tools
Windows Event Logs	Logons, process creation, RDP, privilege use	Event Viewer, EvtxECmd, Chainsaw
UserAssist / RecentDocs	GUI program execution and recently opened files	Registry Explorer, RECmd
ShellBags	Folder access (including deleted/USB/network)	ShellBags Explorer, SBECmd
ShimCache (AppCompatCache)	Files that existed/run on the system	AppCompatCacheParser
AmCache	Executed programs with paths and hashes	AmcacheParser
\$MFT (Master File Table)	Full filesystem timeline (create/modify/access/delete)	MFTECmd, MFTEExplorer, Autopsy
Browser History	Web activity, downloads, attacker research	Hindsight, BrowsingHistoryView
Registry (Run keys, Services, Tasks)	Persistence	Autoruns, RECmd, Registry Explorer

Windows Event Log Analysis

- **Tool: Windows Event Viewer** (or export tools) Key events:
 - **4624** (successful logon — note logon type: 3=network, 10=RDP)
 - **4625** (failed logon)
 - **4688** (process creation — enable command-line logging for details) **Why:** Shows who logged in, when, and what processes ran. Convert times to UTC and filter noise.

NTFS File System Forensics

- **Tool: MFTECmd** or **MFTEExplorer** Example: MFTECmd.exe -f \$MFT --csv timeline.csv **Why:** \$MFT is the central filesystem record. Every file has an entry with timestamps. Use for timelines (filter .exe, Temp folder, etc.).
Timestamps (MACB): M=Modified, A=Accessed, C=Changed (metadata), B=Born (creation). Compare SI (easily altered) vs FN (more reliable) entries.
Small files (< ~700 bytes) can be resident (content stored in MFT — recoverable even if deleted). Additional: \$LogFile and \$UsnJrnl track changes.

Registry & Execution Artifacts

- **Tool: Registry Explorer** or **RECmd** Check UserAssist, RecentDocs, Run keys, etc. **Why:** Reveals GUI executions, opened files, and persistence locations.

Browser Forensics

- **Tool: BrowsingHistoryView** or **Hindsight** **Why:** Aggregates history/downloads across browsers. Shows intent (e.g., research on tools or exfil sites).

Overall Investigation Flow

1. Event logs → authentication and process activity.
2. NTFS/MFT → what changed on disk and when (timelines).
3. Registry artifacts → execution history and persistence.
4. Browser → external activity and intent.
5. Correlate everything into one timeline.